Problem 2: Valorant
4+3 Points

Problem ID: `smoke`
Rank: `1+2`

Introduction

Gigi E. Zee peaked Immortal 3 in Valorant (Episode 2, Act 3) with 276 RR, ranking around 18,000 in the world and needs to reach that again to prove that she isn't washed. Unfortunately, this isn't easy because her teammates always instalock duelists. Ms. Zee, the solo queue omen one-trick, needs to lock in to avoid being flamed in the vc.



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Problem Statement

Omen starts the round with **two smoke charges**. Using a smoke consumes one charge. Once using a smoke, it will take $\mathbf{R}$ seconds for Omen to regain a smoke charge. He can carry a maximum of 2 charges. Smokes recharge sequentially; if both charges are used then the second charge only starts recharging after the first one is restored.

After the round starts, his teammates ask him to use his smoke $\mathbf{N}$ times, at timestamps $\mathbf{A}_i$ for $1 \leq i \leq \mathbf{N}$. The requests are given in chronological order. Omen must have a charge available when a request comes in to fulfill it. Determine whether he can smoke every time his teammates ask for it. Output `YES` if he can fulfill all requests, and `NO` otherwise. Assume that recharges are processed before requests at any given timestamp, i.e. if a charge is set to recharge at the 10 second mark, that charge can be used to fulfill a request at the 10 second mark.

*Note: Templates are available for this problem—and **all other problems in this contest**—in Python, Java, and C++! Find them in the contest.zip provided at the start of the contest. Templates handle input and output for you, so you can just fill out a single function!*

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Input Format

The first line of the input contains a single integer $\mathbf{T}$ denoting the number of test cases.

For each test case:

- The first line contains two space-separated integers $\mathbf{N}$ and $\mathbf{R}$ where:
  - $\mathbf{N}$ denotes the number of smoke requests
  - $\mathbf{R}$ denotes the number of seconds needed for a smoke to recharge
- The second line contains $\mathbf{N}$ space-separated integers $\mathbf{A}_1\ \mathbf{A}_2 \ldots\ \mathbf{A}_N$ denoting the time of each smoke request.

Output Format

For each test case, output `YES` if he can smoke every time his teammates request him too or `NO` if he cannot.

Constraints

$1 \leq \mathbf{T} \leq 100$
Requests are given in chronological order such that $\mathbf{A}_1 < \mathbf{A}_2 < \cdots < \mathbf{A}_N$.

Main Test Set

$1 \leq \mathbf{N} \leq 10^3$
$1 \leq \mathbf{R} \leq 10^3$
$1 \leq \mathbf{A_i} \leq 10^5$

Bonus Test Set

$1 \leq \mathbf{N} \leq 10^5$
$1 \leq \mathbf{R} \leq 10^5$
$1 \leq \mathbf{A_i} \leq 10^9$

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Sample Test Cases

Sample Input Download

```
5
5 10
20 30 40 50 60
7 100
1 2 3 4 5 6 7
4 10
1 2 12 13
8 5
5 10 17 30 36 45 50 56
9 30
20 70 80 90 100 190 210 220 250
```

Sample Output Download

```
YES
NO
NO
YES
NO
```

Sample Explanations

For the first test case, the first smoke is thrown at the 20 second mark, and the smokes recharge every 10 seconds. Each request occurs 10 seconds after the previous, so we will always have 2 smoke charges ready to fulfill the requests. We return `YES`

For the second test case, the first smoke can be thrown at the 1 second mark and the second smoke can be thrown at the 2 second mark, but the next smoke won't recharge until the 101 second mark. Since the following requests happen every second, none of those requests can be fulfilled. We return `NO`

For the third test case, the first and second smoke charges are used at times 1 and 2 respectively. At time 12, the first smoke charge has recharged since eleven seconds have elapsed since using the first charge. However, the next smoke won't recharge until the 22 second mark, so we are unable to fulfill that smoke request. We return `NO`

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